Tony Tran

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EDUCATION

University of Houston, Cullen College of Engineering | Houston, TX

M.S. in Engineering Data Science and AI, GPA: **3.67/4.00** **May 2026**

Thesis: ELASTIC: Efficient Once For All Iterative Search for Object Detection on Microcontrollers

B.S. in Computer Engineering Technology, GPA: **3.84/4.00** **May 2024**

Minor in Technology Leadership and Innovation Management

Engineering in Training (FE/EIT) License in Electrical and Computer (*Texas: 81224*) **June 2024**

EXPERIENCE

University of Houston | Houston, TX

Graduate Research Assistant – Network Autonomous and Intelligent Learning (NAIL) Lab **Dec. 2023 - Present**

- Develop Efficient Neural Architecture Search Algorithms for Edge AI and Embedded Systems
- Reformulated Robot Angle Estimation as a Classification Task (**21 Bins**), Applied **4** Unique Domain Randomization Techniques, Trained Robust DNN (**88% Accuracy**) on **28k** Synthetic Training Images and Validated on **28k** Test Images
- Streamlined Model Development Pipelines through Prompt Engineering and Distributed Training on HPC Clusters and the Lambda Cloud, Reducing End-To-End Experimentation Time by **3x**

Teaching Assistant **Aug. 2024 – Present**

- Lead Weekly Lab Sessions/Office Hours for **30+** Students per Class, Responsible for Grading Homeworks, Tests, Quizzes
- Designed **Six** New Lab Assignments Ranging from Deep Learning Models in PyTorch, Model Compression Techniques (Pruning/Quantization), and CodeGen (Generating C/C++ Code for Edge Models)
- Taught AI in Robotics (SP2026), Embedded Systems (FA2024, SP2025), Sensor Applications (FA2024, SP2025), Introduction to C++ Programming (FA2024)

Research Mentor, UH RE-PST Program **Summer 2025**

- Mentored **Two** End-to-End Embedded Machine Learning Projects, Building Cats vs. Dogs Image Classification and Keyword-Spotting Audio Recognition Demos on **Two** Different Microcontrollers in **<6 Weeks**
- Designed an Intensive AI in Embedded Systems Hands-On Curriculum Containing (**15+** Labs) and Guiding **Two** Undergraduate Researchers with No Prior Experience

SELECTED PROJECTS

ELASTIC – Master's Thesis, Paper, Code | Python, PyTorch (+DDP, AMP), Embedded C **Aug. 2024 - May 2025**

- Designed Lightweight DNNs in under **32 KiB Memory**, Reducing Energy Consumption by **3.5x**, Requiring **1.6x** Less Power Usage, and Lowering Latency by **2.4x** utilizing Proposed Constrained Iterative OFA NAS Framework for Detection
- Deployed Tiny DNNs in under **1/4 MiB Flash** and Achieving **3ms** Inference Time on MAX78000/02 Platforms for Digit and Face Detection Demos
- Map Specialized Neural Network Architectures to Data Memory in Hardware-Accelerated CNN Inference Accelerator

Otto Sorting Bin – Capstone Project, 1st Place, Poster, Video | Jetson Nano, YOLO, OpenCV **Aug. 2023 - May 2024**

- Finetuned YOLOv8 (**90%** Accuracy) Object Detection Model using TACO Detection Subset (**1.5k** Images) and **>500** Custom-Labeled Images for Robust Trash and Recyclable Detection Management on Nvidia Jetson Nano
- Engineered a Low-Cost T-Bot Sorting System and Constrained Object Priority Algorithm (**>85%** Physical Pickup Success) through Mechanical-Algorithmic Co-Design

TrashDet – WACVW2026, Paper, Slides, Video | Python, PyTorch, YOLO, UAV **Nov. 2025 - Dec. 2025**

- Achieved State-Of-The-Art Performance (**19.5 mAP**) on the TACO Dataset (**1.5k** Images), Outperforming Prior Waste Detectors with **2.8x** Less Model Size
- Derived Deployable TinyML Models down to **1.1M** Parameters, Maintaining Real-Time Performance at **37 FPS** for Real Time Waste Detection

Traffic Severity Modeling | Python, Scikit-Learn, Matplotlib, Seaborn, Pandas, NumPy **Aug. 2025 - Dec. 2025**

- Interpreted over **7.7 million** Traffic Records with over **40** Variables and Built Accident Severity Predictive Model (**85%** Accuracy) using Logistic Regression, Linear Discriminant Analysis, SVM, Naïve Bayes, Random Forest, and LightGBM
- Employed PCA and K-Means Clustering to Profile **Four** Data-Driven Crash Categories and Improve Interpretability

ADMM Pruning – Code | Python (+OOP), PyTorch, Matplotlib, NumPy, Pandas, Excel **Aug. 2025 - Dec. 2025**

- Cut Size of Mobile DNNs in **Half**, Retaining **97%** Predictive Performance, and Increasing Accuracy by **3.5x** Post Pruning
- Sped Up Model Convergence by **36%** and Discovered **One** Architecture Failure Mode under High Complexity

1-Lipchitz Layers Beyond Classification – Code | Python (+OOP), PyTorch, Matplotlib **Aug. 2024 - Dec. 2024**

- Extended 1-Lipchitz Robustness Theory to Object Detection, Evaluating **Three** Layer Designs on Tinier-SSD Detectors using the SVHN Detection Dataset (>**40k Images**)
- Benchmarked Robustness, Achieving **10%** Higher Accuracy, and Retaining **85%** Predictive Performance (Compared to **53%** Against Baseline)
- Quantified Cost of Robustness (**6x** Training Time, **20x** Parameters), Informing Production-Scale Trade-Offs

PUBLICATIONS

[1] **T. Tran**, Q. Lin, and B. Hu, "ELASTIC: Efficient Once for All Iterative Search for Object Detection on Microcontrollers", *Accepted at IEEE Transactions on Computers 2026*.

[2] **T. Tran** and B. Hu, "TrashDet: Iterative Neural Architecture Search for Efficient Waste Detection," in Proceedings of the *2026 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW),* Tucson, AZ, USA, Mar. 2026.

[3] R.R. Suganda, **T. Tran**, M. Pan, L. Fan, Q. Lin, and B. Hu, "Distributed Perception Aware Safe Leader Follower System via Control Barriers Methods", *2025 IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, GA, May 2025. (**Acceptance Rate: 38.67%**).

[4] **T. Tran** and B. Hu, "FACETS: Efficient Once-For-All Object Detection via Constrained Iterative Search", *2025 IEEE International Conference on Robotics and Automation (ICRA) Late Breaking Session*, 2025.

AWARDS AND HONORS

- NVIDIA Academic Grant Recipient: Robotics and Edge AI (March **2026**)
- Summa Cum Laude (Awarded by UH **2024**)
- University Honors (Awarded by The Honors College **2024**)
- Dean's List (**8 Semesters 2020-2024**),
- **7x** Engineering Project Awards (**5x 1st** Place, **2x 2nd** Place)

TECHNOLOGIES

- **Languages**: Python, C/C++ (Embedded), Java, HTML, MATLAB, Verilog
- **Machine/Deep Learning**: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LightGBM
- **Computer Vision**: Classification, Object Detection, Segmentation; Ultralytics, Detectron2, OpenCV, Roboflow
- **EdgeAI/TinyML**: Neural Architecture Search, Once-For-All, Pruning, Quantization, ONNX, TFLite, TinyEngine
- **Data Libraries**: SciPy, NumPy, Pandas
- **Visualization**: Matplotlib, Seaborn, Tableau, Canva, Draw.io, Lucid Chart, PowerPoint
- **Hardware**: MAX78000/02 (ARM Cortex-M4F + AI CNN Hardware Accelerator), Jetson Nano (ARM Cortex-A57 + NVIDIA Maxwell GPU), TM4C123GXL (ARM Cortex-M4F), Arduino Uno (ATmega328P), STM32F746G (ARM Cortex-M7)
- **Tools**: Git, Linux, Bash
- **AI Tools**: ChatGPT, Gemini, DeepSeek, Cursor, Claude, NotebookLM